

Fabrication of 3D-culture platform with sandwich architecture for preserving liver-specific functions of hepatocytes using 3D bioprinter

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Abstract: The development of new three-dimensional (3D) cell culture system that maintains the physiologically relevant signals of hepatocytes is essential in drug discovery and tissue engineering research. Conventional two-dimensional (2D) culture yields cell growth, proliferation, and differentiation. However, gene expression and signaling profiles can be different from *in vivo* environment. Here, we report the fabrication of a 3D culture system using an artificial scaffold and our custom-made inkjet 3D bioprinter as a new strategy for studying liver-specific functions of hepatocytes. We built a 3D culture platform for hepatocytes-attachment and formation of cell monolayer by interacting the galactose chain of galactosylated alginate gel (GA-gel) with asialoglycoprotein receptor (ASGPR) of

hepatocytes. The 3D geometrical arrangement of cells was controlled by using 3D bioprinter, and cell polarity was controlled with the galactosylated hydrogels. The fabricated GA-gel was able to successfully promote adhesion of hepatocytes. To observe liver-specific functions and to mimic hepatic cord, an additional parallel layer of hepatocytes was generated using two gel sheets. These results indicated that GA-gel biomimetic matrices can be used as a 3D culture system that could be effective for the engineering of liver tissues. © 2017 Wiley Periodicals, Inc. *J Biomed Mater Res Part A*: 105A: 1583–1592, 2017.

Key Words: tissue engineering, 3D bioprinter, hepatocyte, 3D culture, galactosylated alginate (GA-gel)

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INTRODUCTION

The liver is the largest glandular organ in human body that assists almost every other organ in some facet; it has a complex three-dimensional (3D) architecture with manifold functions. It is composed of several cell types, including hepatocytes and sinusoidal endothelial cells. Hepatocytes form two cell monolayers that separate sinusoidal blood from the canalicular bile. In glandular tissues, cells inhabit in 3D extracellular matrix microenvironment and interact with adjacent cells through biochemical and mechanical cues to develop perfectly organized/polarized complex glands. It is well known that cell-cell and cell-ECM adhesion, interaction and polarity plays a significant role in the regulation of cell morphology, proliferation, differentiation, and viability. It is believed that in order to maintain specificity and glandular

homeostasis, an appropriate 3D cellular/tissue microenvironment is mandatory. The inability of cells to acquire *in vivo*-like architectural design and communication network in conventional two-dimensional (2D) culture assays can retard various critical cellular processes.^{1,2}

Thus, the fabrication of 3D culture microenvironments that mimic complex cellular interactions, *in vivo* tissue organizations, and regulate the cell adhesion/polarity that is lacking in conventional 2D culture monolayers is very significant for making the recovery of 3D structures and maintaining the partial characteristics of the original liver tissue physiology.

Over the last few years, methods for engineering of liver tissues that maintain liver-specific functions and cell polarity have received increasing attention.^{3–10} In general, these efforts have involved the culturing of primary hepatocytes on

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collagen-coated dishes. However, the cultured hepatocytes do not maintain their liver-specific functions in the long run.

Recently, 3D cell cultures have gained a tremendous attention in the biomedical research community with their ability to closely mimic physiological environments. Until now, various porous scaffolds that integrate different biochemical or architectural cues for the construction of 3D microenvironments have been developed.^{11–13} Despite progress in recent years resulting in the generation of 3D culture microenvironments, the better physiological knowledge of the hepatocyte signaling underlying the cellular crosstalk and extracellular matrix microenvironment is restrained. Most of the developed 3D culture platforms suffer from various technical difficulties that limit their potential applications. For example, Nakao et al. developed a novel microfluidic cell culture device that mimics the microscopic structure in liver tissue. Although, the developed microfluidic system shows the capability to regulate the alignment of hepatocytes and form the bile canaliculi,¹⁴ hepatocytes still show difficulties in inducing physiologically relevant signals and formation of a 3D environment that fully mimics liver tissue.

One emerging strategy for addressing these shortcomings in liver tissue engineering research is 3D printing. We have ever developed a custom made inkjet 3D bioprinter which has the capability to fabricate fine 3D complicated structures by printing live cells and biomaterials.^{15–18} The inkjet technology possess many great potential applications due to its wide configurability which allows the ejection of micro-sized droplets. Our 3D bioprinter is supplemented with a piezoelectric inkjet head system which can easily eject ink materials such as living cells and hydrogels without generating excess heat.

In our previous work, we successfully applied our custom-made bioprinting system to generate various 3D gel structures using alginate hydrogels as bioink.^{16–18} Although, alginate hydrogel fabrication procedure is very easy but its poor cell adhesive characteristics limit its long time potentiality in maintaining tissue-specific physiological and pathophysiological activities.¹⁹ It is well known that cell-cell crosstalk between hepatocytes and their surrounding cells are essential for liver development and maintenance of liver functions. Thus, alginate hydrogel is not a suitable material for liver tissue engineering research because alginate is incapable of regulating the polarity of hepatocytes.

As an important step forward, we focused on applying the scaffold with galactosylated materials such as lactose-carrying styrene polymer (PVLA) and galactosylated alginate (GA). PVLA and GA are the specific polymers that show the ability to adhere to hepatocyte through the interaction between the asialoglycoprotein receptor (ASGPR) of hepatocyte and the galactose moieties of PVLA and GA.^{20–25}

Towards that end, this study aimed to develop the sandwiched culture system in which 3D geometrical arrangements of cells were controlled by using 3D bioprinter. The polarity of hepatocytes was controlled with the galactosylated hydrogels. As anticipated, the fabricated biomimetic GA-gel structure provide a confined 3D microenvironment

exhibiting significant adhesion property toward hepatocytes. To mimic hepatic cord and to resemble the natural *milieu* of hepatocytes more clearly, a double layer of hepatocytes using two gel sheets was fabricated. We also checked the polarity of hepatocytes on the artificial matrix having galactose chains. In addition, the ability of the sandwich culture system to maintain liver-specific functions was also evaluated. These results indicated that our 3D culture system can be used as a promising tool for biomimetic cell culture, studying cellular responses, and obtaining predictive data for clinically useful drugs.

MATERIALS AND METHODS

Materials

Fetal bovine serum, penicillin and streptomycin, buffered saline (PBS), *N*-hydroxysuccinimide (NHS), gelatin, trypsin/EDTA, mouse anti- β -actin, and trypan blue were purchased from Sigma-Aldrich (MO). Sodium alginate solution (MW 216 g/mol) was obtained from Kyosei Pharmaceutical Co., Ltd. (Hokkaido, Japan). Barium chloride dihydrate ($\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$), ethylenediamine, dimethyl sulfoxide (DMSO), paraformaldehyde (PFA), and *N,N,N',N'*-tetramethylethylenediamine (TEMED) were purchased from Wako Pure Chemical Industries (Osaka, Japan). Water soluble carbodiimide hydrochloride (WSC) was purchased from Dojindo Molecular Technologies (Kumamoto, Japan). Lactobionic acid (LA) was purchased from Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan). Acti-stain 488 phalloidin was purchased from Cytoskeleton Inc. (CO, USA). Calcium chloride (CaCl_2) and blocking one were purchased from NACALAI TESQUE, Inc. (Kyoto, Japan). Williams' E (WE; Invitrogen) medium was purchased from Invitrogen (CA, USA). Rabbit anti-ASGPR, goat anti-albumin, and goat anti-MRP2 antibodies were purchased from Santa Cruz Biotechnology Inc. (CA, USA). PVLA was prepared according to the method described previously,²⁶ and dissolved in water.

3D bioprinter

The 3D bioprinter has been described in detail, previously.¹⁸ As shown in Figure 1(A,B), the printer is composed of a main printer unit, inkjet control unit, and personal computer unit for control. Its size was designed for use in a bioclean bench. In addition, a computer-controlled elevator stage was also used to support the 3D bioprinting system. Figure 1(C) shows a schematic diagram for the fabrication of 3D gel structures by using the previous method and using the new improved 3D bioprinter approach. By adding the computer-controlled elevator stage it is possible to control a Z-axis finely, and layer-by-layer fabrication can be performed more precisely.

Fabrication of gel structure

The alginate gel structure was fabricated according to previously described methods.¹⁸ Alginate gel is useful in fabricating 3D structure by crosslinking through calcium ion and carboxyl group of sodium alginate. The viscosity of the synthesized GA solution was confirmed to be suitable for ejecting from the 3D bioprinter. The synthesized GA polymer was prepared at a final concentration of 12 mg/mL. The GA solution was ejected into a 20 mg/mL $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ solution

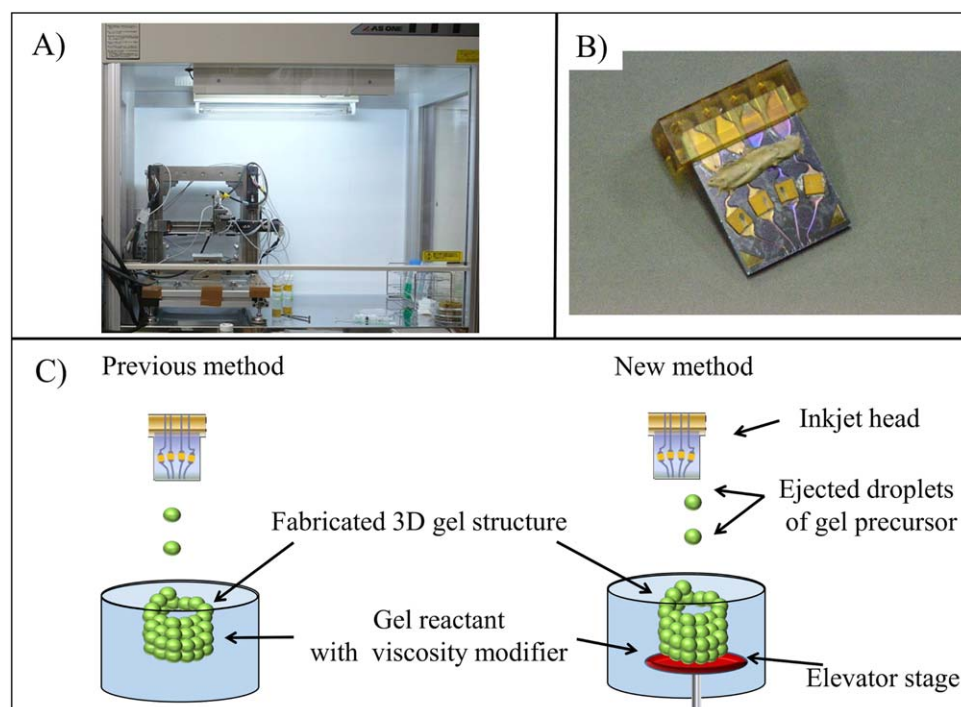


FIGURE 1 The 3D bioprinter and the scheme of the gel structure fabricated method. A: General view of the 3D bioprinter. B: Inkjet nozzle head. C: The gel structure fabrication of previous method and new method.

for which the viscosity had been adjusted with viscosity-modifying materials such as poly(vinyl alcohol) and hyaluronan. The GA or Alg-gel sheet was fabricated by using the PDMS mold to fabricate the sandwich culture device.²⁷ Briefly, GA or Alg solution was added to the PDMS mold. Then, GA or Alg solution in the well of the PDMS mold was immersed in a 20 mg/mL $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ solution and instantaneously gelled. The length and breadth of the well in the PDMS mold were approximately $1.5 \text{ cm} \times 1.5 \text{ cm}$. Finally, a gel sheet of sodium alginate and GA was fabricated by using the 3D bioprinter.

Preparation of galactosylated alginate (GA)

The protocol for synthesizing the GA polymer is shown schematically in Figure 2. The GA polymer was prepared from the reaction of sodium alginate and monoamine terminated lactobionic lactone (L-NH₂) by a method similar to that described previously.²⁴ Briefly, lactobionic lactone was prepared by dehydrating lactobionic acid (LA). The lactobionic lactone was then reacted with 30-fold ethylenediamine in anhydrous dimethyl sulfoxide (DMSO) for 24 h. After this reaction, L-NH₂ was precipitated using chloroform, and the precipitated L-NH₂ was vacuum dried. The sodium alginate was dissolved in TEMED/HCl buffer (pH 4.7). This solution was activated by adding NHS and EDC and allowing the reaction to proceed for 1 h. After this activation, L-NH₂ was added and the solution was reacted for an additional 24 h. The reacted solution was purified by using a dialysis tube (12,000–14,000 MWCO) against distilled water for 24 h, and freeze-dried by using a lyophilizer.

Galactose assay

To confirm the galactose activity, the GA solution was analyzed by using a Galactose Assay Kit (Wako, Japan). Briefly, 50 μL each 0.5 wt % sample was added to 96-well plates. Next, 50 μL of the mixture of galactose assay solution was added to each well plate and incubated at 37°C for 30 min. Absorbance was measured at 570 nm by using a microplate reader.

Preparation of the extracellular matrix-coated dish

Matrix coating was carried out as previously described.^{28–32} Briefly, 3 mg/mL collagen, 100 $\mu\text{g}/\text{mL}$ PVLA, and 0.5 mg/mL or 5 mg/mL GA solution were added to a 24-well plate at 37°C and incubated overnight. After incubation, the nonabsorbed solution was removed and washed with PBS three times. Then, 1 mg/mL bovine serum albumin solution was added to the plates and the plates were left at 37°C for 30 min to prevent nonspecific adherence of cells. After incubation, the nonabsorbed solution was removed and washed with PBS three times and preserved at 37°C until use in the culture of hepatocytes.

Isolation and culture of hepatocytes

The primary hepatocytes were isolated from the liver tissue of male 6- to 8-week-old ICR 12 mice (SLC, Japan) by using a two-step *in situ* collagenase perfusion method.³² Under anesthetization with pentobarbital, the livers of the mice were preperfused with Hank's Buffered Salt Solution (HBSS) containing EGTA (0.19 g/L) and glucose (0.98 g/L) to remove the blood after the cannulation in the portal

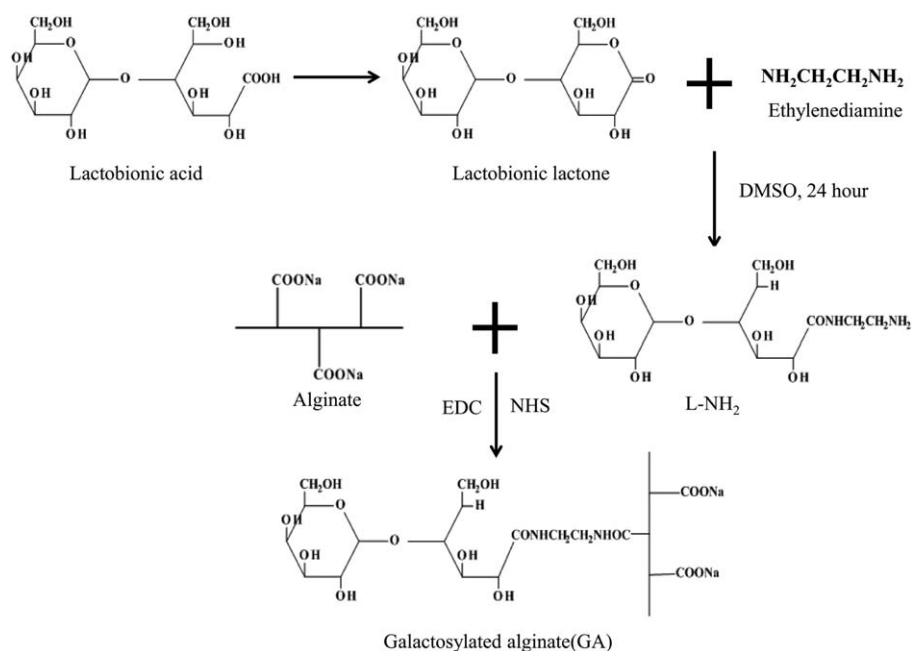


FIGURE 2. The scheme of GA synthesis.

vein. Following perfusion with 0.015% collagenase in HBSS the liver was eviscerated, and the tissues were dispersed in Williams' E medium including FBS by using blade mincing for isolating cells. After dispersing, the cells were filtered through a 100 μm nylon cell strainer and centrifuged twice for 2 min at $50 \times g$ to remove non-parenchymal cells. After washing the cells, the hepatocytes were centrifuged in 40% Percoll solution for 10 min at $90 \times g$ to remove the dead cells. The cell viability was confirmed to be $>85\%$ by using the trypan blue exclusion method.

Sandwich culture

These primary hepatocytes were cultured on GA gel sheets for 8 h. As shown schematically in Figure 3, to achieve a sandwich architecture, the primary hepatocytes on GA gel sheets were inverted, and placed on another GA gel sheet also containing primary hepatocytes. After making the sandwich structure, doubled-layer gel sheets were cultured at 37°C for 1 day.

Western-blot analysis

The protein of hepatocytes was extracted with lysis buffer and 1% protease inhibitor cocktail. The cell lysate was centrifuged

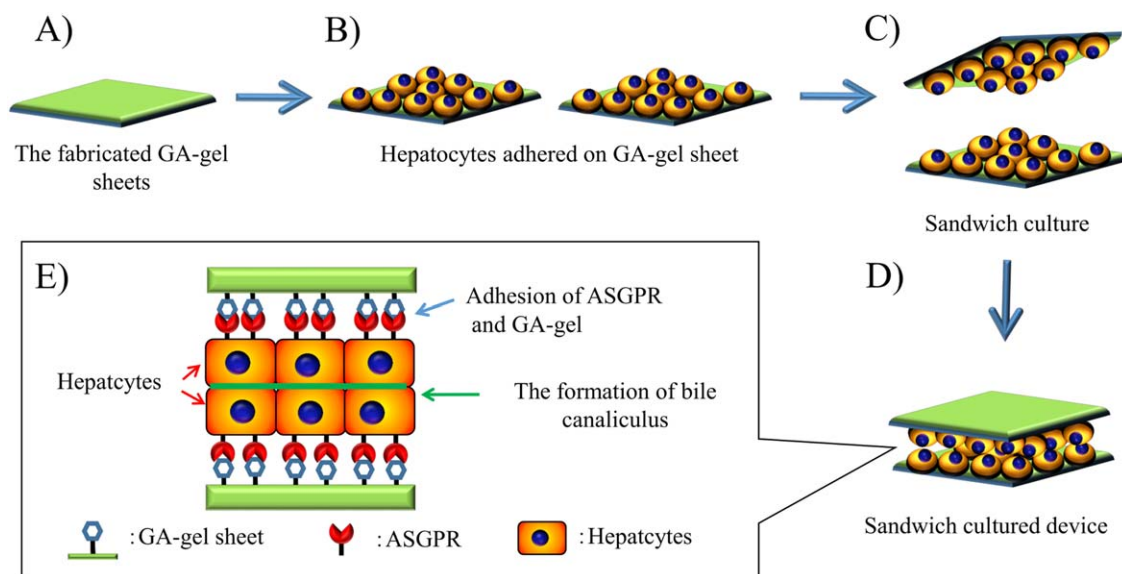


FIGURE 3. The scheme of the sandwich culture method. A: The fabricated GA-gel sheets by using 3D bioprinter. B: The hepatocytes seeded and adhered on GA-gel sheets. C: The scheme of sandwich culture by using two separate layers of hepatocytes on GA-gel sheets, (D) the fabricated sandwich culture device.

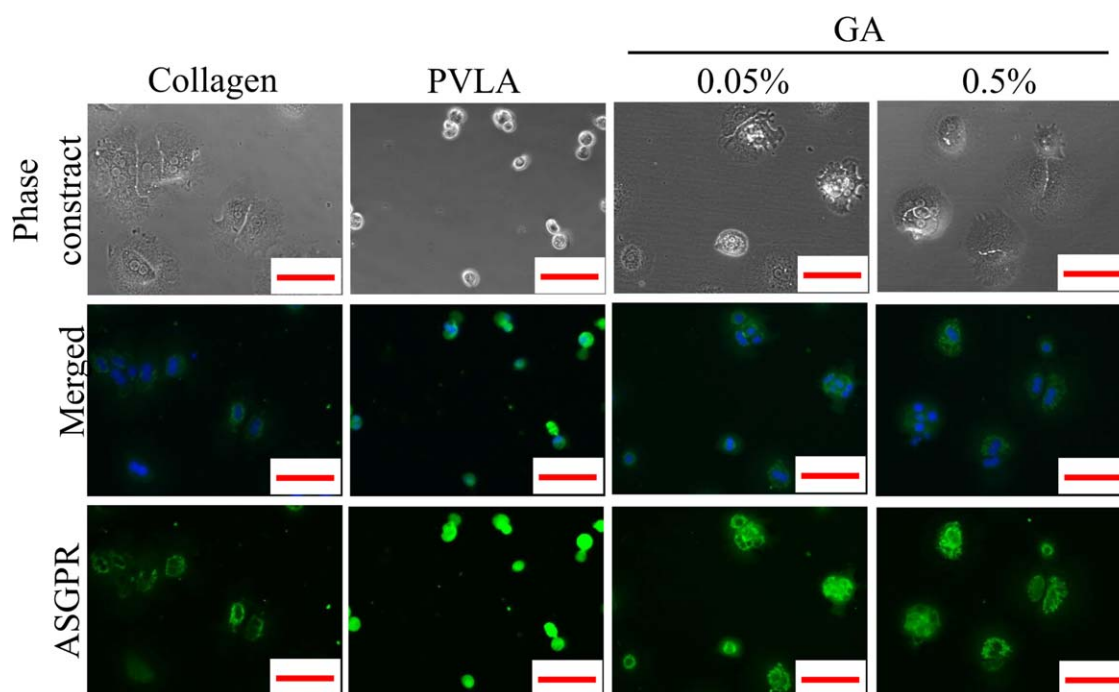


FIGURE 4. Morphology and expression of ASGPR of hepatocytes on dishes coated with different matrices. A–D: Phase contrast observation of hepatocytes on different matrices such as collagen or PVLA or GA. E–H: A merged image of nuclei (blue) and ASGPR (green) in hepatocytes on dishes coated with different matrices, as visualized using the immunofluorescent staining method. I–L: An immunofluorescent staining image of ASGPR expression in hepatocytes on dishes coated with different matrices. Blue: nucleus; green: ASGPR; the scale bar represents 100 μm .

at $15,000 \times g$ for 30 min at 4°C and the supernatant was collected. The protein of hepatocytes was adjusted in the sample buffer, and separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) on 10% polyacrylamide gels. The separated samples were transferred onto a poly(vinylidene difluoride) (PVDF) membrane. The transferred membrane was treated by blocking at 4°C overnight by using blocking one. After blocking, the membrane was incubated with primary antibodies (rabbit anti-ASGPR, mouse anti- β -actin) and incubated overnight at 4°C . After incubation, the membrane was incubated with HRP conjugated secondary antibodies (1:10,000 dilutions) for 2 h at room temperature, and HRP activity was assayed with Immobilon Western detection reagents.

Histological analysis

The cultured primary hepatocytes were fixed in 4% paraformaldehyde (PFA) solution for 60 min. After fixing, sample was washed with tris-buffered saline (TBS) including 50 mM CaCl_2 solution three times. The sample was then embedded in 7 wt % gelatin solution. After gelling, the samples were immersed in 10, 15, and 20 wt % sucrose solution at intervals of 12 h. These gel structures were frozen by using OCT compound, sections were made by a cryostat, and stained with hematoxylin and eosin.

Immunohistochemical analysis

The cultured primary hepatocytes on several types of ECM-coated dishes were fixed in 4% paraformaldehyde (PFA)

solution for 30 min. After fixing, these samples were washed with PBS three times and permeabilized with 0.03 wt % Triton-X solution for 5 min. After drying on the thin sections, the cells and/or sections were washed with PBS three times and blocked with 1% BSA solution for 30 min. After blocking, these samples were reacted with rabbit anti-ASGPR, goat anti-albumin or goat anti-Mrp2 as the primary antibody at 4°C overnight. They were washed three times using PBS, and incubated with secondary antibodies conjugated with Alexa Fluorophore. The F-actin of hepatocytes was stained with Alexa Fluor 488 phalloidin. The nuclei of all samples were stained with Hoechst 33342.

RESULTS

Morphology and expression of ASGPR of hepatocytes on dishes coated with different matrices

To assess morphology and expression of asialoglycoprotein receptor of hepatocytes on culture dishes coated with different matrices, we seeded the hepatocytes on surfaces modified with collagen, PVLA, and GA matrix, respectively. After 4 h, it was observed that hepatocytes readily attached to and promoted spreading shapes on dishes precoated with collagen. Whereas, hepatocytes cultivated on PVLA and GA-coated surfaces showed attachment with round shapes but did not promote spreading. In order to gain an insight into the expression level of asialoglycoprotein receptor in hepatocytes, we cultured the hepatocytes on collagen, PVLA and GA matrices and incubated for 1 day. The results presented in Figure 4 demonstrated that the expression level of

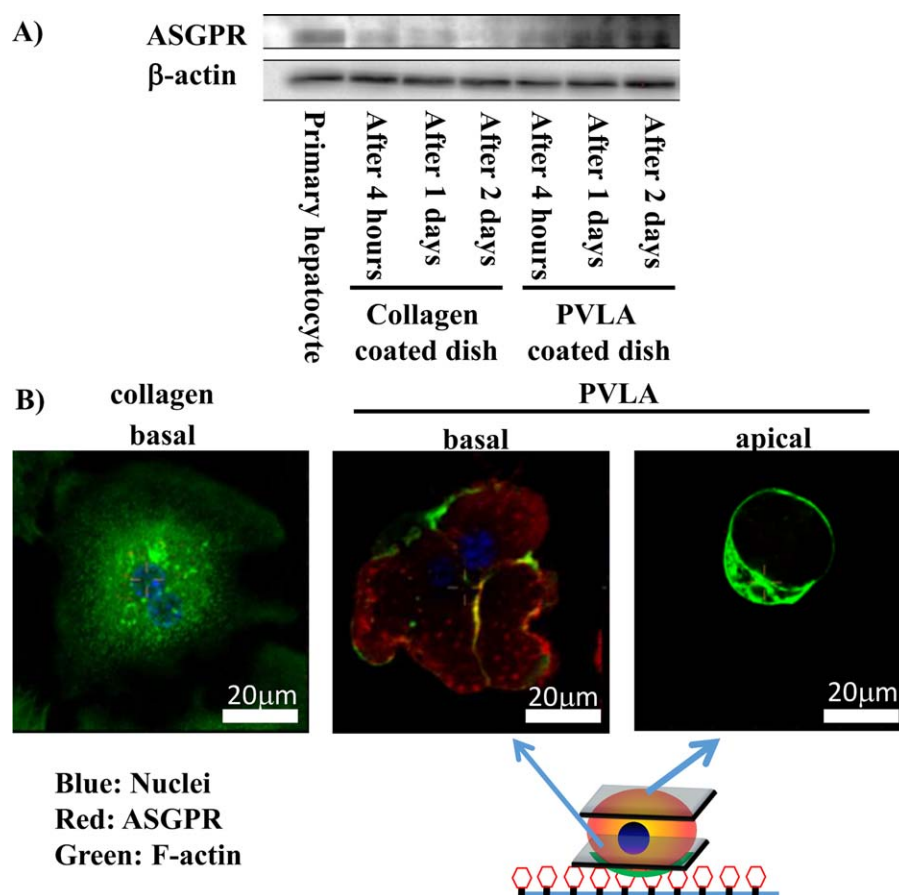


FIGURE 5. Western-blot analysis and immunohistochemical staining of ASGPR. A: The ASGPR expression of primary hepatocytes on collagen or PVLA-coated dish was confirmed Western-blot analysis. The protein level was normalized by using β -actin expression. B: The comparison with the ASGPR of hepatocytes on collagen or PVLA coated dish.

asialoglycoprotein receptor in hepatocytes cultured on the PVLA- or GA-coated dishes was significantly higher than the cells maintained on the collagen-coated dishes.

Relationship between the distribution of ASGPR in hepatocytes and duration of culture

At 4 h of culture, ASGPR existed in the cultured hepatocytes on both collagen- and PVLA-coated dishes, but as the culture time increased further the ASGPR expression level began to differ between the two matrices. On the collagen-coated dishes the expression gradually decreased, while the hepatocytes on the PVLA-coated dishes began to express ASGPR at increasingly higher levels until 2 days of culture (Fig. 5A). The distribution of ASGPR was very different between the hepatocytes on the collagen-coated dishes and the PVLA-coated ones. On the former ASGPR expression was very weak, and it was not possible to see the orientation in the cells, while on the latter the distribution of ASGR appeared to be limited to the basal site of the cells (Fig. 5B). These findings indicated that the matrix having galactose chain in PVLA induced the expression of ASGPR and regulated its distribution in the hepatocyte.

Adherence rate of cultured hepatocytes on alg-gel and GA-gel sheets

Both the alginate-gel and GA-gel sheets were successfully fabricated by using the 3D bioprinter. The hepatocytes adhered and survived on both types of gel sheets [Fig. 6(A,B)], but the number of cells on the GA-gel sheet was greater than that on the alginate-gel sheet [Fig. 6(C)]. This result indicated that the GA-gel sheet was a suitable scaffold for the adhesion of hepatocytes.

Synthesis of GA and fabrication of a complex 3D gel structure using a 3D bioprinter

To confirm galactose activity, the synthesized GA polymer was checked by galactose assay. The GA polymer exhibited a higher level of galactose activity than the sodium alginate polymer [Fig. 7(A)]. The viscosity of the 1.2% GA solution was the same as that of the 0.8% sodium alginate solution, and this was the usual viscosity required for the 3D bioprinter to eject [Fig. 7(B)]. Thus, bioprinting strategy was applied to fabricate complex 3D gel structure using a 1.2% GA-gel as ink material. For printing process, Bitmap image data of a desired structure was selected and then a 3D computer model of the 3D structure was reconstructed from the

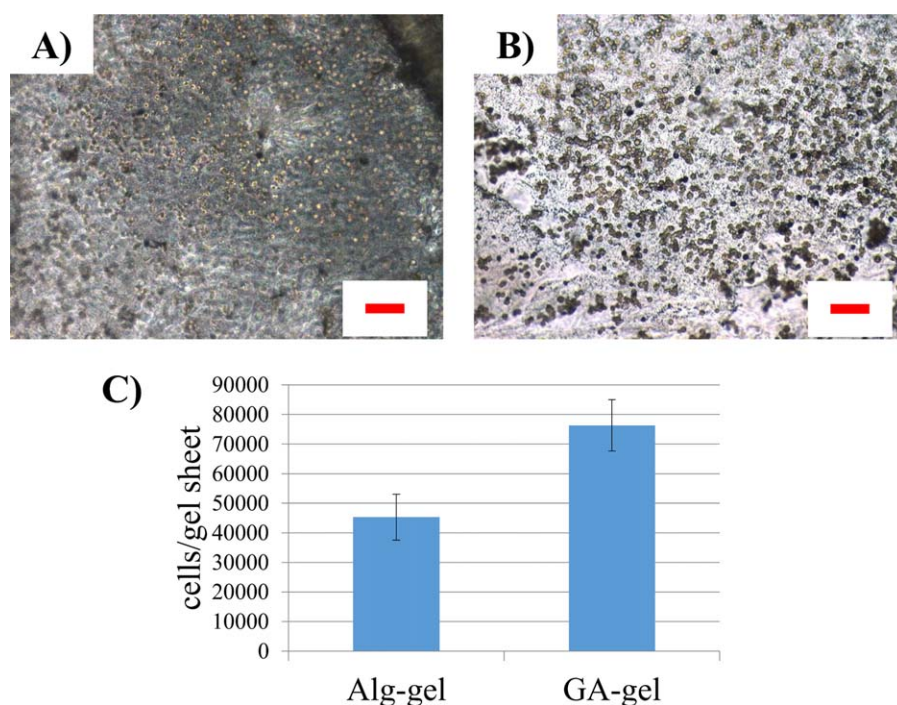


FIGURE 6 Adhesion of hepatocytes on Alg-gel sheets and GA-gel sheets. A: Phase contrast image of hepatocytes on Alg-gel sheets and (B) hepatocytes on GA-gel sheets. C: Amount of hepatocytes on Alg-gel and GA-gel sheets. This result is presented as the mean of three independent experiments. The scale bar represents 100 μm .

selected bitmap images. The results presented in Figure 7(C,D) revealed that 3D gel structure was successfully fabricated in a layer by layer fashion.

Fabrication of the sandwich culture device

The sandwich culture device was fabricated by using two layers of the hepatocytes on the GA-gel sheets (Fig. 3). Staining with hematoxylin and eosin confirmed that the cells were arranged in a monolayer in the monolayer culture device, and in the sandwich culture device the cells were attached via cell to cell contact with double layer formation [Fig. 8(A,B)]. Immunohistochemical analysis showed that the cells in these devices

expressed liver-specific markers such as albumin, MRP2, and ASGPR [Fig. 8(C-J)]. In case of ASGPR, the expression in the monolayer device was located exclusively in the cytoplasm, while in the sandwich device the distribution of this marker was limited to the cell membrane [Fig. 8(H,I)]. These results indicated that the sandwich culture device could arrange the two layers of cells and maintain their liver-specific functions.

DISCUSSION

It is imperative to construct *in vitro* 3D cell culture systems for liver tissues that can more realistically resemble *in vivo*

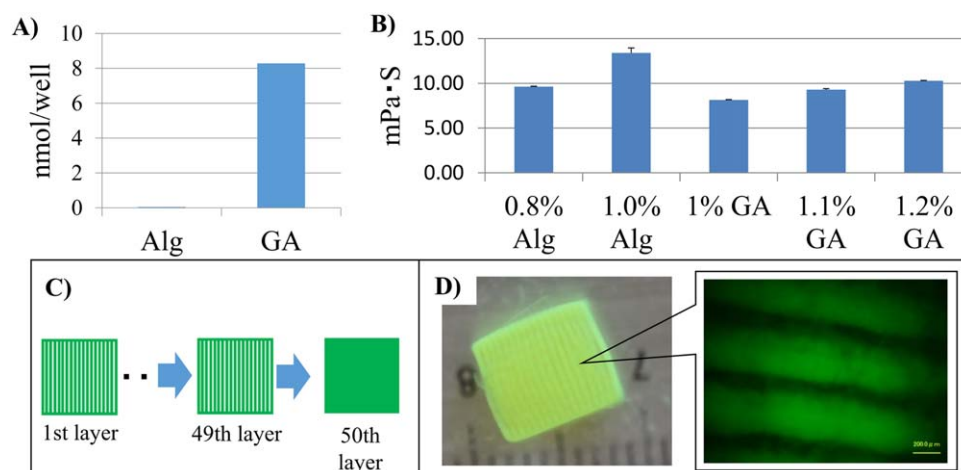


FIGURE 7 Evaluation of synthesized GA polymer and the fabrication of 3D gel structure by using a 3D bioprinter. A: The galactose assay of sodium alginate (Alg) and galactosylated alginate (GA). B: The viscosity of Alg and GA solution. C: The fabrication process with multiple images. D: The fabricated 3D gel structure.

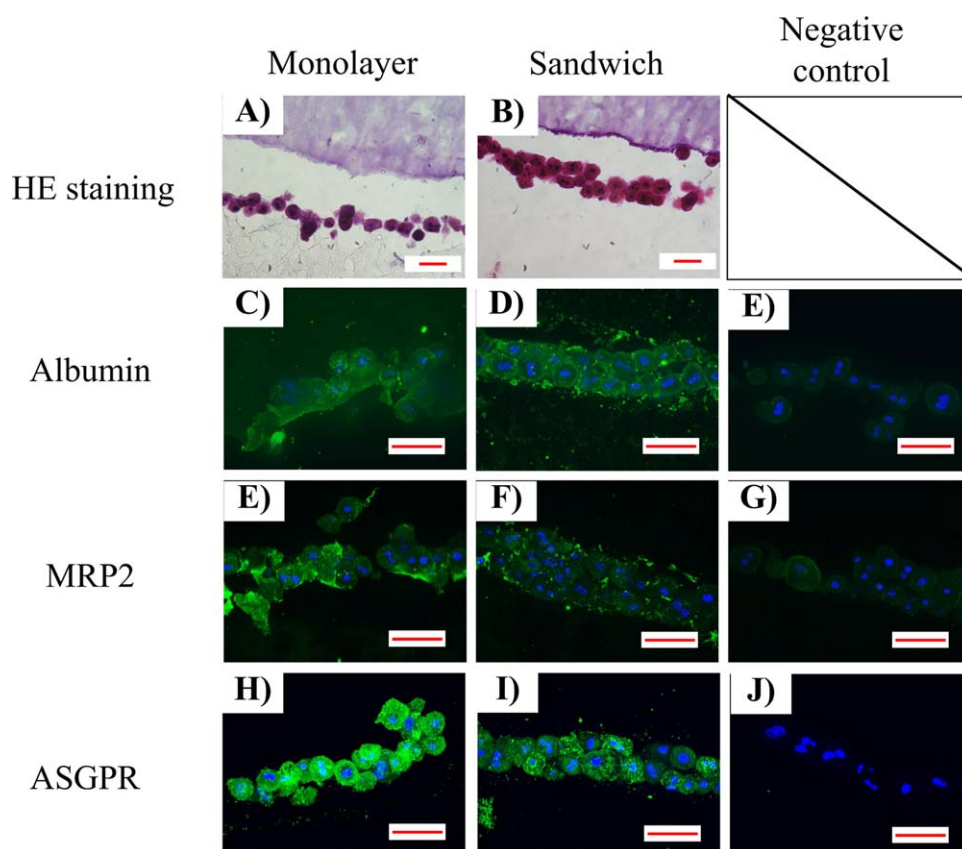


FIGURE 8. Histological and immunofluorescent staining. Hematoxylin and eosin staining of monolayer (A) and sandwich (B). Immunocytochemical analysis of albumin (C–E), MRP2 (F–H), and ASGPR (I–K). The scale bar represents 50 μm .

architecture of tissue and provide physiologically relevant information. It is well known that the expressions of specific cell membrane proteins of hepatocytes are localized *in vivo* at the vessel side and the bile canaliculi side, respectively. However, there are no effective techniques to control the expressions of such specific membrane proteins in case of 3D condition. To determine hepatocytes grown in our 3D culture system offered a benefit over other 2D and 3D cultures for identifying the liver specific properties of cells with greater selectivity for hepatocytes, we established a novel 3D culture system and optimized the parameters to grow two layers of the hepatocytes on the GA-gel sheets in a sandwich architecture. Our main objective was to create an artificial extracellular matrix (ECM)-like microenvironment with the ability to control the polarity of hepatocytes. Several groups have already reported that the hepatocytes cultured on different types of ECM-coated dishes expressed some liver-specific functions such as albumin production and cytochrome P450 (CYP) enzyme activity.

Recently, Nakamura and Ijima reported that the hepatocytes cultured on decellularized liver film expressed higher levels of liver-specific activities than those cultured on collagen film.⁶ This suggested that the ECM in the liver played an important role in the liver functions. The ECM of liver tissue is not only composed of collagen, but also many other kinds of proteins as typified by laminin, fibronectin, proteoglycan, and so on. Similarly, Adachi et al. reported that the

mRNA expression of the CYP gene in hepatocytes cultured on lactose-carrying styrene polymer-coated dishes was significantly higher compared to collagen-coated dishes.³⁰ In addition, Hoshiba et al. also reported that the hepatocytes grown on PVLA-coated dish were capable to maintain several liver-specific functions.³¹

PVLA is one of the specific polymers that adheres hepatocytes through the interaction between the asialoglycoprotein receptor (ASGPR) of hepatocytes and the galactose moieties of PVLA polymer. But it can also easily attach to the surface of the culture dish by hydrophobic interaction.²⁸ We considered that PVLA, which included galactose chains, would be suitable as an artificial matrix to control the polarity of hepatocytes. Indeed, the expression of ASGPR of hepatocytes cultured on the collagen-coated dishes decreased with time, in contrast to that of hepatocytes on the PVLA-coated dishes (Fig. 5). In PVLA-coated dishes, ASGPR was expected to be confined to only apical portion of hepatocytes. The GA chain in PVLA was believed to induce the polarity of hepatocytes after culturing on the PVLA. Unfortunately, it is difficult to use PVLA to fabricate 3D materials, because alginate hydrogel is a hydrophilic polymer.³³

Therefore, we synthesized galactosylated alginate (GA) polymer to fabricate a 3D gel structure because it contains a sufficient number of galactose chains, which can permit the easy interaction of galactose chain of galactosylated alginate gel (GA-gel) with asialoglycoprotein receptor (ASGPR)

of hepatocytes. The concentration of galactose chains in the synthesized GA polymer was measured using a galactose assay kit, and the results showed that the concentration was higher in the synthesized GA polymer than in the sodium alginate alone. To determine the availability of primary and secondary amines in GA solution, ninhydrin assay was performed. As expected, no amines were detected in the GA solution. It has already been reported by Yang et al. that GA gel is suitable substrate to maintain hepatocytes viability or liver-specific functions.²⁴ Thus, the hepatocytes cultured on GA-coated dishes preserved their physiologically relevant characteristics. These results indicated that GA polymer which was synthesized by using sodium alginate and L-NH₂ possesses enough properties for culturing the hepatocytes.

The gel-formation potential of the GA polymer was checked by using the 3D bioprinter. In general, the carboxyl group of sodium alginate was allowed to react with BaCl₂ solution which resulted in the fabrication of the gel structures. The synthesis process of GA polymer was carried out by covalent coupling chemistry between the carboxyl group of sodium alginate and the amine group of L-NH₂. It was expected that the GA polymer is lacking the capability for the excellent formation of gel structures. Contrary to expectations, the 12 mg/mL GA solution was able to fabricate a complex 3D gel structures by using the 3D bioprinter.

For comparative studies, the hepatocytes were seeded on both the Alg-gel and GA-gel sheets. Interestingly, hepatocytes maintained on the GA-gel sheet showed excellent adhesion characteristics. Whereas, hepatocytes maintained on alginate-gel sheet showed very less adhesion.

It is widely believed that the alginate material cannot be absorbed by several types of hydrophilic proteins present in the ECM. For example, Li et al. reported that hepatocytes exhibited almost no adherence to either Alg-gel or Alg-coated dishes.¹⁹ However, the scaffold based on the alginate hydrogels has good advantage for cell adhesion as biomaterials. Our results demonstrate that using the galactose-modified alginate hydrogels, the hepatocytes can specifically adhere by interaction of between cell and GA-gels without other ECM. Compared to Alg-gel sheet, the GA-gel sheet was found to be more suitable alternative for maintaining the adhesive properties of hepatocytes.

Finally, a sandwich architecture was successfully fabricated using GA-gel sheets adhered with hepatocytes. The hepatocytes in the sandwich culture environment using GA-gel sheets were arranged in two layers and preserved their liver-specific functions as in the actual 3D environment. The results showed a clear spherical shape of hepatocytes, their connections with the neighboring cells were resembling with the native liver tissue environment. Moreover, the hepatocytes of the sandwich culture device also expressed liver-specific proteins and receptors such as albumin, MPR2 and ASGPR. In particular, a part of the MPR2 in hepatocytes on sandwich culture device was observed, and it indicated the polarity of hepatocytes. This data shows that the sandwich culture device could regulate the polarity of hepatocytes by improving the morphology of GA-gel structures.

This data is expected that the sandwich culture device may mimic part of the hepatic cord structure as well.

In conclusion, the overall focus of this work was to develop a sandwich 3D culture system for hepatocytes culture with the capability to mimic the liver specific functions. We prepared galactosylated alginate gel (GA-gel)-based 3D culture system using galactose chain chemistry of galactosylated alginate gel (GA-gel) for the interaction with asialoglycoprotein receptor (ASGPR) as the basis for an *in vitro* 3D culture platform for hepatocytes-attachment and cell monolayer formation. The resulting sandwich structure was found to be biocompatible for in 3D hepatocyte cell cultures. We showed that the 3D geometrical arrangement of cells can be controlled by our custom made 3D bioprinter, and cell polarity can be controlled with the galactosylated hydrogels. These properties could be matched to those of natural liver tissue environments. We also determined that an optimal density at which hepatocytes could be seeded in the gel sheets, and observed hepatocytes adhesion property. These results clearly suggested that GA-gel biomimetic matrices can be used as a favorable 3D culture system that could be effective for the engineering of liver tissues, studying cellular responses and pharmaceutical research analysis

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